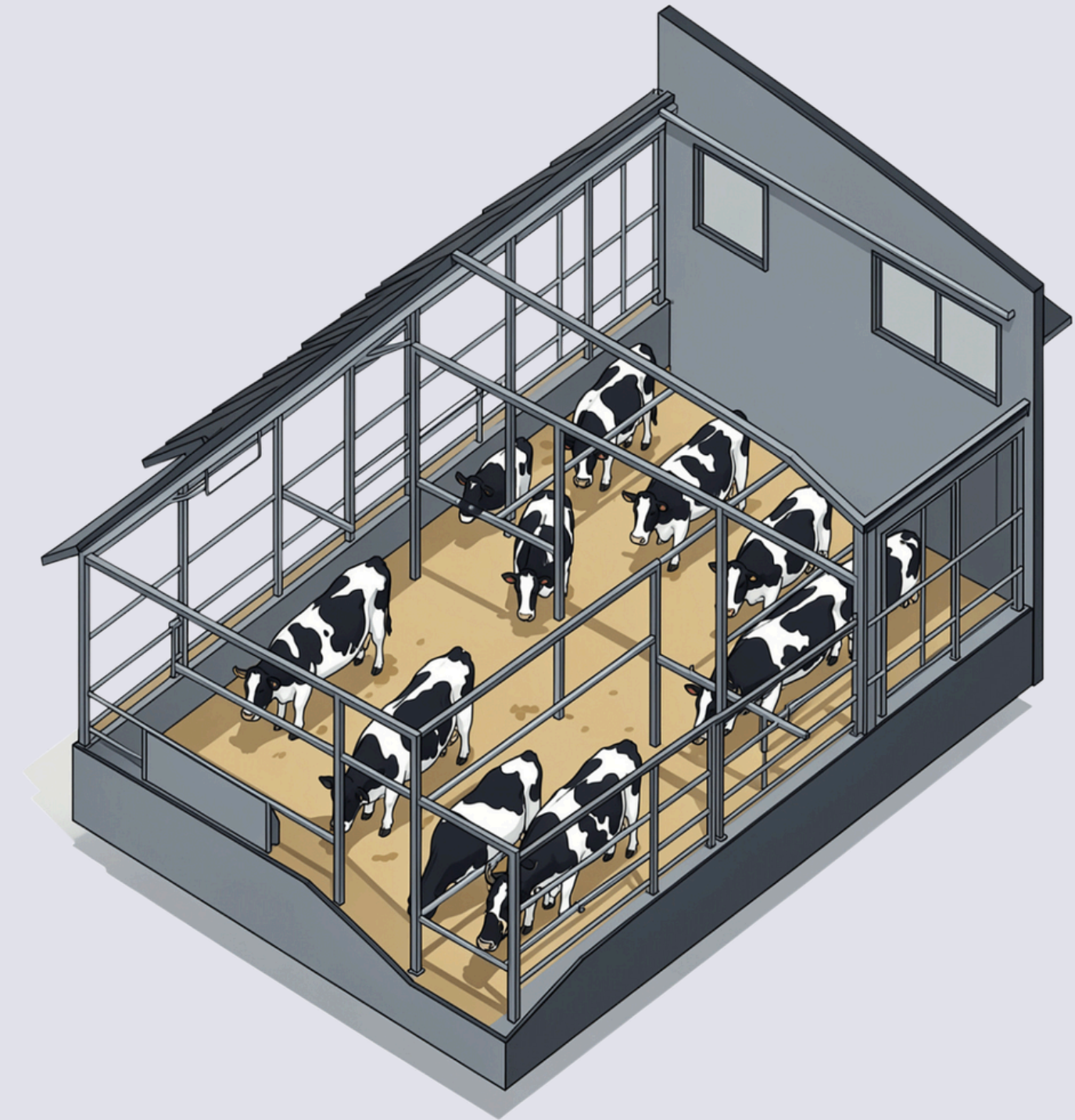


Dairy Farm Herd & Milk Collection Management System

Presentation-II: Database Architecture, Relational Schema & SQL Query Pack

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Faculty Mentor: Dr. Kiran Mayee Adavala | **Technology:** MySQL Community Server 8.0.46



Project Overview & Relational Architecture

Transforming Traditional Dairy Registers into an ACID-Compliant Enterprise RDBMS

14

InnoDB Tables

Strongly-typed relations across operations, veterinary, inventory, and billing domains

3NF

Normalization

Formally normalized schema eliminating all update, insert, and delete anomalies

4

Active Triggers

Automated inventory guards and invoice reconciliation logic

700

Production Rows

Exactly 50 realistic, referentially-intact records per table

Problem Statement & Objectives



LEGACY VS. MODERN

The Legacy Problem

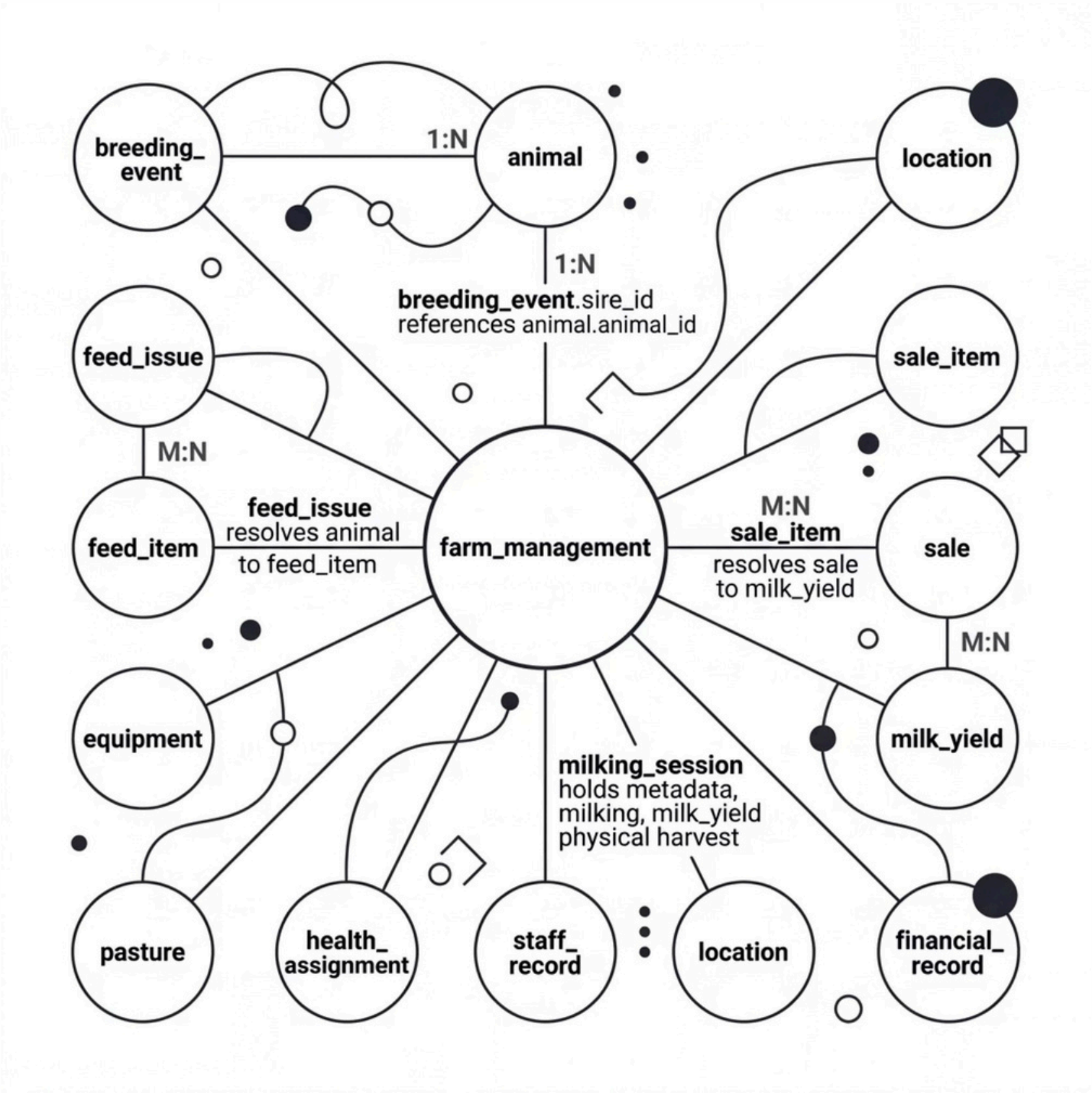
- Logbook fragmentation with no centralized record-keeping
- Lack of attribution across herd health and milking events
- Financial mismatch between bulk shipments and cooperative payments

DBMS Objectives

- Referential integrity enforced with ON DELETE CASCADE
- Formal decomposition into Third Normal Form (3NF)
- Transactional ACID safety on MySQL 8.0 InnoDB engine

Modernizing Dairy Operations through Relational Discipline

Enhanced Entity-Relationship (EER) Architecture



14 Entities

Unary, binary, and associative mappings across the entire operational domain.

Recursive Lineage

breeding_event.sire_id references animal.animal_id in a unary 1:N relationship.

M:N Associative Bridges

- feed_issue resolves animal ↔ feed_item
- sale_item resolves sale ↔ milk_yield

2-Table Harvest Normalization

milking_session holds operational metadata; milk_yield holds physical harvest quantities.

Authoritative Relational Schema & Constraints

	Core Entities breed (PK: breed_id) animal (PK: animal_id, FK: breed_id → breed [RESTRICT])
	Health & Breeding health_visit, vaccination, breeding_event FK: animal_id [CASCADE]; sire_id [self-ref]
	Inventory & Feed feed_item (PK: feed_id) feed_issue (PK: issue_id, FK: feed_id, animal_id)
	Milk Operations milking_session → milk_yield → quality_test All FKs: [CASCADE]
	Sales & Payments buyer, sale [RESTRICT], sale_item (composite PK), payment [CASCADE]

Normalization Justification:

1NF to 3NF



First Normal Form (1NF)

All attributes store **atomic scalar values**. Repeating multivalued attributes decomposed into individual child tuples.

2

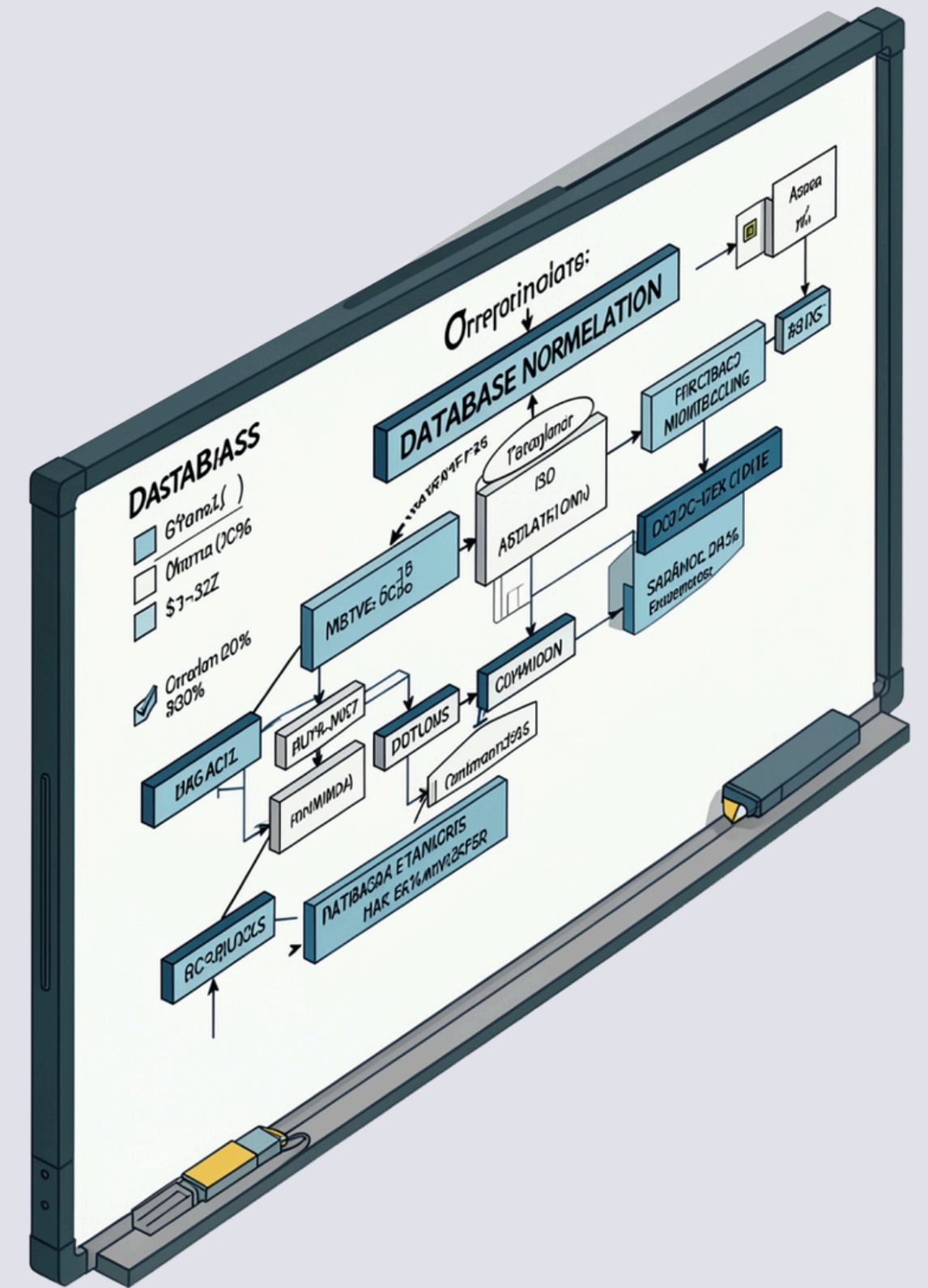
Second Normal Form (2NF)

In `sale_item(sale_id, yield_id)`, `quantity_sold` and `unit_price` depend strictly on the **entire composite key**.

3

Third Normal Form (3NF)

Eliminates transitive dependencies such as $\text{animal_id} \rightarrow \text{breed_id} \rightarrow \text{breed_name}$ by decomposing into separate breed relation.



Business Automation: 4 Active Triggers & 1 View

Trigger Details

`before_feed_issue`

Validates `feed_item.available_quantity`; raises SIGNAL SQLSTATE 45000 if dispensation exceeds physical inventory.

`after_feed_issue`

Atomically decrements `feed_item.available_quantity` upon each confirmed cattle feed issuance.

`after_sale_item`

Computes $(\text{quantity_sold} \times \text{unit_price})$ and increments `sale.total_amount` automatically.

`animal_milk_summary` View

Aggregates `total_sessions`, `total_litres`, and `avg_yield_per_session` for real-time dashboards.

Validate Inventory

Decrement Stock

Compute Sale

Aggregate Summary

Comprehensive SQL Query Pack: 30+ Production Queries



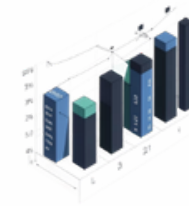
Filtering & Aggregation

- `DISTINCT` & `BETWEEN` for range queries
- `LIKE` & `IN` for pattern matching
- `GROUP BY` with `HAVING` exceeding 20.0 L thresholds



Advanced Joins & Subqueries

- Multi-table `INNER JOINS` across operational domain
- `LEFT OUTER JOINS` for uninspected lots
- Correlated subqueries for yield anomaly detection



Window Functions

- `ROW_NUMBER()` ranking animals by cumulative milk production
- `RANK()` for quality tier classification



Set Operations

- `UNION` for consolidated health & vaccination reports
- `EXCEPT` for animals never recorded in sales

Key Performance Indicators & Insights



Herd Productivity

- Total milk yield per animal & seasonal trends
- Lactation cycle efficiency metrics

Inventory Optimization

- Feed consumption rates per head
- Waste minimization via automated issue tracking

Financial Reconciliation

- Sale-to-payment matching & buyer settlement accuracy
- Revenue per animal over reporting period

Health & Breeding

- Vaccination compliance rates & vet visit frequency
- Breeding success metrics & lineage tracking

Conclusion & Future Enhancements

Current Achievement

Enterprise-grade RDBMS replacing fragmented logbooks with a fully **ACID-compliant, 3NF-normalized** architecture across 14 InnoDB tables.

Scalability

Partitioning strategies for **multi-year historical data**; read replicas architected for reporting workloads without impacting OLTP performance.

Future Roadmap

Real-time IoT sensor integration for milking metrics; predictive analytics for breeding outcomes; mobile app for field operations.

